



ROOT



MATLAB Geek

Summer Bootcamp | 2026

WiseDuck



Code is poetry written for
machines!

Presented by



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Let's Code!



The Wanderer



EXAMPLE 1,0 | Damping Type

$$a \ddot{I} + b \dot{I} + c I = 0$$

$$\alpha = b/2a$$

$$\omega_0 = \sqrt{c/a}$$

$$at^2 + bt + c = 0$$

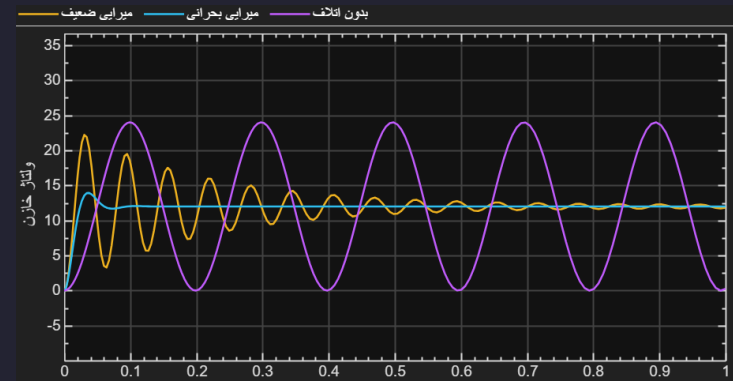
$$t = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

$\alpha = 0 \rightarrow$ Undamped (بدون اتلاف)

$\alpha < \omega_0 \rightarrow$ Underdamped (میرایی ضعیف)

$\alpha = \omega_0 \rightarrow$ Critically Damped (میرایی بحرانی)

$\alpha > \omega_0 \rightarrow$ Overdamped (میرایی شدید)





EXAMPLE 1,1 | Infinite Resistor Network

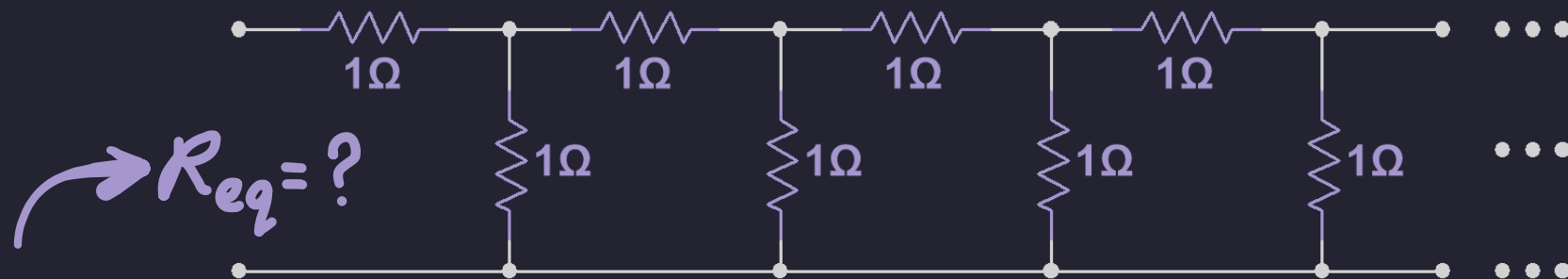
$$R_{eq} = 1 + (1 \parallel (1 + 1 \parallel (1 + 1 \parallel (1 + \dots$$

$$R_{eq} = 1 + (1 \parallel R_{eq}) \Rightarrow R_{eq} = 1 + \frac{R_{eq}}{1 + R_{eq}}$$

$$\Rightarrow R_{eq}^2 - R_{eq} - 1 = 0$$

$$\Rightarrow R_{eq} = \frac{1 + \sqrt{5}}{2} = 1.618034\Omega \text{ (Golden Ratio 🏆)}$$

$$R_{eq} = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \dots}}}$$





EXAMPLE 1,2 | Bisection

$$f(x) = e^x - \cos(x) - 1$$

$$f(x) = 0; x = ?$$

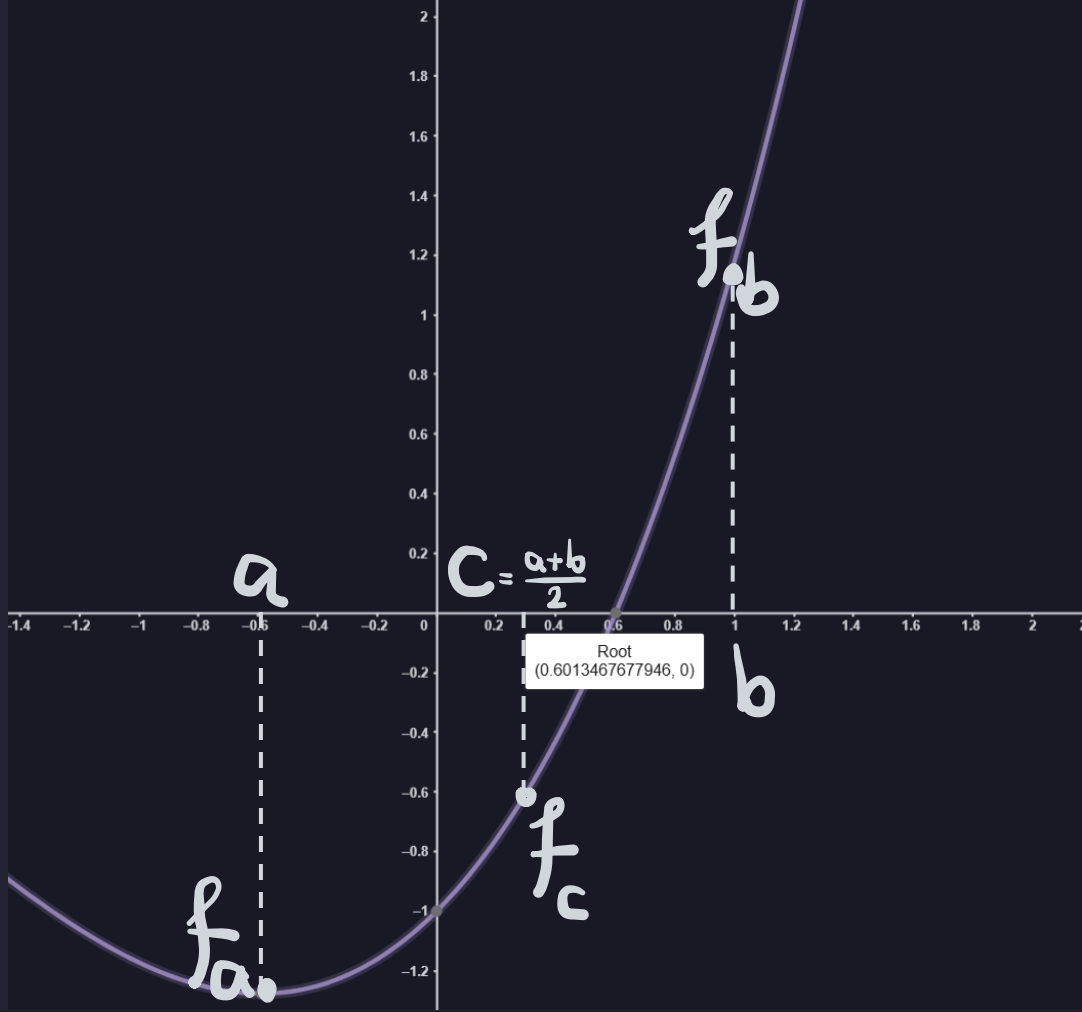
$$c = \frac{a+b}{2}$$

loop

$$f_a \cdot f_c < 0 \rightarrow b := c$$

$$f_b \cdot f_c < 0 \rightarrow a := c$$

$$\Rightarrow x = c$$







The End!



THANKS!

Do you have any questions?



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